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Date: 9/26

PRELAB:

Q1. Consider the Verilog code in section 3.0. Briefly explain how the **always @** structure works.

Continually executes whats inside it when one of the inputs change

Q2. Write the Verilog code for *lab4step1*. Use the example code given in Section 3.0 and make the necessary changes.

```
module farmers_problem(A, F, C, G, W);  
    input F, C, G, W;  
    output A;  
    reg A;  
  
    always @(F, C, G, W)  
    begin  
        case({F, C, G, W})  
            4'b0000: A = 'b0;  
            4'b0001: A = 'b0;  
            4'b0010: A = 'b0;  
            4'b0011: A = 'b1;  
            4'b0100: A = 'b0;  
            4'b0101: A = 'b0;  
            4'b0110: A = 'b1;  
            4'b0111: A = 'b1;  
            4'b1000: A = 'b1;  
            4'b1001: A = 'b1;  
            4'b1010: A = 'b0;  
            4'b1011: A = 'b0;  
            4'b1100: A = 'b1;  
            4'b1101: A = 'b0;  
            4'b1110: A = 'b0;  
            4'b1111: A = 'b0;  
        endcase  
    end  
endmodule
```

Q3. Read Section 4.0 and fill in the Truth Table for *lab4step2*.

Inputs				Outputs		
M	T	H	P	E	F	AC
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	1	0	1
0	1	0	0	0	0	0
0	1	0	1	1	0	1
0	1	1	0	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	1	0	0
1	1	0	0	0	0	0

Lab 4 Answer Sheet

1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	1	1	1

Pre-lab TA Initials: _____

LAB:

3.0 Use the hardware results to fill in the truth table for *lab4step1*.

Farmer	Cabbage	Goat	Wolf	Alarm
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0

1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

ModelSim results demonstrate correct code. TA Initials: _____

4.0 Demonstrate ModelSim results for correct A/C code. TA Initials: _____